

Environmental, energy and economic analysis of a biomass supply chain based on a poplar short rotation coppice in Spain



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ABSTRACT

This paper provides an environmental, energy and economic analysis of a biomass supply chain based on an experimental poplar plantation grown as a Short Rotation Coppice in Southern Spain. The investigation was based on Life Cycle Analysis (LCA) methodology, considering two harvesting technologies and including a sensitivity analysis on the effect of transportation distance. The results show that harvesting was the most significant life cycle phase both in terms of environmental damage and economic costs, primarily due to the strong energy demand of the machinery employed in these activities. Transportation had a comparatively lower environmental and economic significance than harvesting, although the effect of this operation increased rapidly with distance. Agricultural operations related to the use of fertilizers and herbicides were found to have a limited effect on the economics of the system but they had a notable impact on certain midpoint categories relating to human and environmental toxicity, eutrophication and acid emissions. Normalized LCA results point to agricultural land occupation as the most affected environmental category of the system. The bioenergy supply system was highly energy efficient with a Cumulative Energy Demand and a Net Energy Ratio determined at 29.1 MJ/GJ and 34.0, respectively. Poplar collection by chip harvesting was less energy intensive and was more benign to the environment. The results also show a linear relationship between biomass transportation distance and impact values in all midpoint LCA indicators. Biomass collection by chip harvesting was more cost-effective than bale harvesting, while transportation costs were decisive in the economic viability of the system. However, in the conditions assumed, current wood chip prices are too low to make this project economically viable in Spain. This conclusion may be different for other locations that have reduced rental costs for land, lower irrigation needs or improved biomass production yields.

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1. Introduction

Industrialized countries are developing strategies to reduce their dependence on fossil fuels and to favor the sustainable utilization of renewable energy resources for the production of power, thermal energy and transportation fuels. Modern bioenergy is expected to play a key role in this transformation of the energy sector (European Commission, 2009; IDAE, 2010; IEA, 2012). At present, most of the solid biomass employed for energy purposes is generated as a by-product of the forest, agriculture and wood-processing

sectors. However, residual materials alone are unlikely to be sufficient to meet the augmented demand that is foreseen in the coming years. Short Rotation Coppice (SRC) plantations are being investigated as a means to supply large amounts of high quality lignocellulosic biomass in a controlled, sustainable and cost effective manner (ENCROP, 2009; Halford et al., 2010). In particular, the production of poplar is attracting scientific and commercial interest in many areas of the world including the Mediterranean basin (Cañellas et al., 2012; Sixto et al., 2010; Testa et al., 2014), Northern Europe (Nielsen et al., 2014; Verlinden et al., 2013) and North America (Elferjani et al., 2014; Jassal et al., 2013). This species is known for its strong resistance to frost, its capacity to re-sprout from its stump and its fast growth rates when cultivated under adequate climatic and soil conditions.

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Despite its high water requirements, industrial production of poplar is particularly well established in Southern and Northern Spain, primarily for veneer production (Tolosana et al., 2011). Conventional poplar plantations are managed in rotation periods that usually span 10–15 years, with tree densities typically ranging between 350 and 700 trees/ha and stump removal at the end of each rotation (Barrio-Anta et al., 2008; Cañellas et al., 2012; Perez-Cruzado et al., 2014). Higher densities (>15,000 trees/ha) and faster rotation cycles have been reported to increase biomass yields to values in excess of 35 t_{DM}/ha·yr (Sixto et al., 2007; Testa et al., 2014). Recent publications state that poplar plantations have the potential to supply up to 1.0% of Spain's energy demand by the year 2020 (Perez-Cruzado et al., 2014).

Despite the renewable nature of this energy resource, agricultural activities necessarily incur environmental costs that need to be quantified and managed accordingly. LCA has been widely used to analyze the sustainability of bioenergy systems (Cherubini and Stromman, 2011) and of poplar SRC plantations for the production of power (Butnar et al., 2010; Gasol et al., 2009a,b; Gonzalez-Garcia et al., 2014) and ethanol (Gonzalez-Garcia et al., 2014). These publications have all been produced by the same research group and are based on the analysis of a specific SRC in Soria (Northern Spain) planted at a density of 10,000 trees/ha and cultivated in three consecutive 5 year cycles. Net greenhouse emissions were reported to range between 1.6 and 2.0 g CO₂ eq per MJ of biomass energy (Gasol et al., 2009a,b; Gonzalez-Garcia et al., 2014). This impact was attributable primarily to the use of synthetic fertilizers and also to the consumption of diesel in the transportation of the biomass to the energy plant. Net Energy Efficiencies were calculated to range between 25 and 26 kJ/MJ for heat production and 81–83 kJ/MJ for power generation.

The economics of biomass supply chains based on SRC have also been investigated in a number of recent publications. The results are highly variable due to differences in biomass yields (which depends on many parameters like soil type, climatology, water availability, species and clone, etc.), land rental costs, wood chip market prices and the availability of public subsidies (Ericsson et al., 2009; Hauk et al., 2014; Testa et al., 2014). Gasol et al. (2010) calculated the supply cost to be between 18.7 and 24.0 €/t_{DM} for poplar wood chips intended for power generation in Southern Spain. Significantly higher costs were reported by Manzone et al. (2014) (45–115 €/t_{DM}) for poplar wood chips produced in Italy. Ericsson et al. (2009) estimated the costs of biomass generation by SRC in Europe to range between 4 and 5 €/GJ, equivalent to 72–90 €/t_{DM}.

This paper provides up-to-date information about the environmental and energy performance of a biomass supply chain based on a high density poplar SRC plantation in Southern Spain. The economic viability of the system is also calculated using the same scope, which enables a quantitative evaluation of the symmetry (or antisymmetry) in the economic/environmental significance of each one of the elements in the life cycle of the system. The analysis includes an assessment of two harvesting technologies (round baling vs. chip harvesting) and incorporates a sensitivity analysis of the effect of transportation distance from the plantation plot to the energy plant (10–100 km). This range of distances has been regarded as economically acceptable in publications analyzing the effect of transportation on the viability of bioenergy projects (Hamelinck et al., 2005; Ranta and Rinne, 2006).

2. Methodology

2.1. Description of the poplar energy crop

Unless otherwise indicated, the technical and economic information employed in this analysis is based on a 2.0 ha plot located in

the Genil Valley (Granada, Spain) (+37° 12' 13.72", 3° 42' 46.30") planted with *Populus x canadensis* clone AF2 at a density of 13,333 trees/ha. The plot was established in 2009 and it is managed as a SRC assuming four consecutive 3 year rotations followed by stump removal at the end of year 12. Experimental data was available for the first rotation and data for subsequent rotations was estimated according to information published in the literature (as described in each case). The soil is classified as clayey silt, the plot is located at an altitude of 602 m, the average rainfall (2009–2011) is 479 mm and the average temperature (2009–2011) is 15.7 °C. Additional information about the cultivation system may be found in Cañellas et al. (2012) and in Perez-Cruzado et al. (2014).

As shown in Table 1, biomass generation at the end of the first rotation period was 51.3 t_{DM}/ha, corresponding to an annual yield of 17.1 t_{DM}/ha × yr. This value is comparable to those reported by other authors for high density poplar plantations in Europe (Butnar et al., 2010; Gasol et al., 2009a,b; Gruenewald et al., 2007). A 10% increase in biomass production was assumed in the second rotation period (compared to first rotation) and an additional 5% production increase was considered in rotations three and four (Faasch and Patenaude, 2012).

2.2. Life Cycle Analysis (LCA) methodology

The LCA analysis was performed according to standard methodology ISO 14040-14044:2006 considering the following phases: Soil preparation and conditioning; Cultivation; Harvesting and transportation; and Soil recovery. ReCiPe 2009 Europe H (Midpoint and Endpoint) v1.09 was used to calculate aggregated impacts on selected environmental categories. The Cumulative Energy Demand (CED) of the system was calculated using CED v1.08 (Hischier and Weidema, 2010; Jassal et al., 2013). The Net Energy Ratio (NER) represents the fraction between the net energy generated and consumed by the system. SimaPro v8.0 software was used to build the models and perform calculations.

2.2.1. Goal and scope

The goals of this investigation are to quantify and evaluate:

- the environmental and energy performance of the experimental biomass supply chain,
- the effect of two harvesting technologies (bale harvesting vs chip harvesting),
- the effect of transportation distance (from 10 to 100 km) and,
- the economic viability of the bioenergy system in its different scenarios.

The Functional Unit (FU) employed in this analysis was 1.0 GJ of net energy in the form of poplar wood chips delivered at the energy plant.

2.2.2. Technical and environmental inventory data

Table 2 provides a summary of all the processes considered in the LCA of the biomass supply chain. Full inventories are supplied

Table 1
Populus x canadensis mass and energy yields in Southern Spain.

	Rotation				12 year average
	1st ^a	2nd ^b	3rd ^b	4th ^b	
Biomass yield (t _{DM} /ha yr)	17.1	18.8	19.8	19.8	18.9
Energy productivity (GJ/ha yr)	312	343	360	360	344

^a experimental data for 1st rotation.

^b estimations for 2nd, 3rd and 4th rotations.

Table 2Technical characteristics and economic inventory (in €₂₀₁₄) of the poplar SRC in Southern Spain.

LCA phases	Processes	Cycle year												Direct cost/income (€)		
		1	2	3	4	5	6	7	8	9	10	11	12	€ ha ⁻¹ (# applications)	€·fresh tonne ⁻¹ (% moisture)	€ t _{DM} ⁻¹
Soil preparation and conditioning	Subsoiling	X												110	–	–
	Plowing	X												40	–	–
	NPK-Fertilizing	X												217	–	–
	Planting	X												650	–	–
Cultivation	Irrigation	X	X	X	X	X	X	X	X	X	X	X	X	297 (×12)	–	–
	Mechanical weeding	X	X			X			X			X		40 (×5)	–	–
	Chemical weeding (pre-emergence)	X			X			X			X			51 (×4)	–	–
	Chemical weeding (post-emergence)			X	X									84 (×2)	–	–
	N-Fertilizing					X		X			X			142 (×3)	–	–
	Land rental	X	X	X	X	X	X	X	X	X	X	X	X	757 (×12)	–	–
	Sub-soiling												X	450	–	–
	Harvesting			X			X		X				X	1242 (×4)	16.5 (25)	22.0
Soil Recovery	Bale transportation (25 km)			X			X		X				X	594 (×4)	7.9 (25)	10.5
	Bale chipping at plant			X			X		X				X	264 (×4)	3.5 (25)	4.7
	Chip harvesting			X			X		X				X	1178 (×4)	12.5 (40)	20.8
	Chip transportation (25 km)			X			X		X				X	466 (×4)	4.9 (40)	8.2
Average incomes	Bale harvesting			X			X		X				X	4525 (×4)	60 (25)	80.0
	Chip harvesting			X			X		X				X	4242 (×4)	45 (40)	75.0

as [supplementary information](#) to this paper. *Soil preparation and conditioning* include the following processes: sub-soiling, plowing, planting and NPK fertilizing. Diffuse emissions of phosphates/nitrates into natural waters associated with the use of NPK fertilizers were calculated according to [Powers \(2005\)](#) and [Cherubini et al. \(2009\)](#), and diffuse emissions of N₂O, CH₄ and NH₃ into the atmosphere were calculated according to guidelines published by [IPCC \(2006\)](#).

The *cultivation* phase includes the following processes: N fertilizing, herbicide treatment, mechanical weed control and irrigation. Diffuse emissions from N fertilizing were calculated considering the same assumptions described above. Oxyfluorfen is a pre-emergency herbicide and was applied in the first year of each rotation. Glyphosate is a post-emergency herbicide and was applied only in the 2nd and 3rd years of the first rotation. Diffuse emissions of these chemicals into the environment were calculated using the PestLCI 2.0 tool ([Dijkman et al., 2012](#)). These results are included in the [supplementary information](#). Irrigation was done by flooding assuming a yearly water consumption of 2.767 m³/ha in the first rotation and a 10% increase in each one of the subsequent rotation periods ([Perez-Cruzado et al., 2014](#)). Weeding was done twice in the first rotation and once in the other three rotations using a hoeing roller.

The *harvesting* phase includes all the collection of the biomass in the fields at the end of each rotation period and its delivery to the energy plant in the form of wood chips. The scenario referred to as *bale harvesting* involves felling and round baling of the biomass using a commercial Biobaler™. The bales remain in the fields for 1–3 months where moisture content is reduced from 50 wt% in the fresh biomass to 25 wt% (always wet matter basis). Transportation to the energy plant is performed using a 38 t lorry (70 m³ volume capacity) where the bales are reduced to wood chips. The scenario referred to as *chip harvesting*, involves biomass collection using a Claas Jaguar chip forager. In this case, the biomass is transported to the energy plant in the form of wood chips containing 40 wt% moisture using 40 t lorries (90 m³ volume capacity).

Transportation requirements were calculated taking into consideration the volume capacity of the lorries, the bulk density of the biomass and its moisture content at the time of transportation.

The number of trips was determined as the fraction between the biomass produced at the end of each rotation period and the transportation capacity of the vehicle assuming maximum volume loading. A correction factor was incorporated to compensate for the lower vehicle loading assumed in Ecoinvent v3.0 database.

The *soil recovery* phase involved rising the plot by sub-soiling at the end of the last rotation period. This is a common practice in commercial poplar plantations and results in the destruction of the root system, hampering the incorporation of the organic matter into the soil ([Sixto et al., 2010](#)). Hence, carbon sequestration has not been considered in the LCA analysis.

The following elements were left out of the scope of the analysis:

- Production of poplar cuttings
- Carbon sequestration by plant roots
- Energy conversion of biomass at the energy plant
- Indirect changes in land use.

The electricity mix considered in this evaluation was that of Spain in 2013 ([REE, 2013](#)). Direct fixation of atmospheric carbon by the trees has not been considered in the quantification of Greenhouse Gas (GHG) emissions. Background inventory data was obtained from Ecoinvent v3.0 database. Energy and material input values (kg/ha), electricity use and the specific characteristics of the machinery (size, weight, capacity) were adapted from Ecoinvent v3.0 considering the information obtained from the experimental plot in Granada (Spain).

2.3. Economic inventory and methodology

Table 2 describes current direct production costs and revenues, which were gathered during spring 2014 from local companies and associations, consultants, distributors and Spain's Environment and Agriculture Ministry (MAGRAMA). Machinery and manpower inputs were based on data from the experimental plot in Granada (Spain) and were calculated according to the methodology described by [Ackerman et al. \(2011\)](#), [Savoie et al. \(2012\)](#) and [Spinelli et al. \(2009\)](#). Hourly requirements of the Biobaler™ and the Claas

Jaguar forager were adapted from [Savoie et al. \(2012\)](#) and [Spinelli et al. \(2009\)](#), respectively. Transportation costs were calculated using the online simulator produced by the Basque Regional Government (Spain) ([Gobierno Vasco, 2014](#)). Indirect costs, associated with the coordination and supervision of subcontracted activities were assumed to account for 5% of all direct costs. Revenues were obtained from selling the wood chips at the end of each rotation period. These were calculated assuming a present market price of fresh wood chips produced by bale harvesting (25 wt% moisture) of 60 €/t and of wood chips produced by chip harvesting (40 wt% moisture) of 45€/t. Net Present Values (NPV) and Internal Rates of Return (IRR) were calculated for different harvesting and transportation scenarios. Cash inflows and outflows were actualized for 2021 (end of cultivation period) assuming a 5.0% annual discount rate and 0.0% inflation rate for the duration of the project.

3. Results and discussion

3.1. Environmental analysis of the bioenergy supply system: base case scenario

[Table 3](#) shows the characterized impacts on selected environmental categories for the scenario referred to as base case, which assumes bale harvesting of the poplar plantation at the end of each rotation period, transportation of the bales to the energy plant located 25 km away from the plot and chipping of the bales for the production of wood chips. The overall impact of the biomass supply system on the climate change category was 2.33 kg CO₂ eq/GJ. The life cycle phases contributing the most to this category are harvesting (57.3%) and cultivation (30.9%).

The harvesting phase (which also includes transportation of the biomass to the energy plant) also showed the highest contribution in most other environmental categories such as fossil depletion, photochemical oxidant formation, human toxicity, and freshwater and marine ecotoxicity (above 60% in all cases). The significance of the cultivation phase was particularly remarkable on four environmental categories: agricultural land occupation, water depletion and marine eutrophication (above 70% in all cases). The other two life cycle phases considered (soil conditioning and soil recovery), primarily the latter, had a less remarkable contribution to the environmental performance of the system.

Direct water use for irrigation amounts to 9.34 m³/GJ. This represents 74.7% of the value estimated for the water depletion category. The high direct water input required by the poplar trees may pose a limitation to the deployment of these plantations in arid areas where this resource may be scarce. Most of the remaining impact in the water depletion category is attributable to the life cycle of fuels employed in the cultivation and harvesting phases.

[Fig. 1](#) describes the environmental significance of specific processes in the harvesting and cultivation phases. Regarding the cultivation phase (left), the results suggest that *N* fertilizing generated the highest impacts on most environmental categories, followed by *glyphosate* and *oxyfluorfen*. Mechanical weeding had a very limited significance on the environmental performance of the system.

Regarding the harvesting phase, the highest contribution in most environmental categories is attributable to the collection of the biomass from the fields using the Biobaler™. The transportation process has a significantly lower contribution (typically between 10 and 20 %). Chipping of the bales at the energy plant has a marked effect on certain categories (water depletion 64.7%, marine and freshwater ecotoxicity 25.9 and 29.1%, and fresh water eutrophication 25.8%) due primarily to the high intensity of this activity in terms of power consumption.

The use of phytosanitary and fertilizing products is usually considered a critical issue in the sustainability of agricultural systems. [Table 4](#) describes the environmental impacts generated by different stages in the life cycle of these chemicals when used in the poplar plantation. The results suggest that the industrial production phase is responsible for most of the environmental impacts associated with the use of herbicides. Diffuse emissions into the environment (air, water and soil) have a negligible significance on the environmental performance of the system. The application phase contributes primarily to climate change due to the use of fossil fuels in the phase. The same is observed in the case of *N* fertilizing, except for terrestrial acidification where most of the impact is associated with the formation and diffusion in the atmosphere of NH₃ and N₂O.

[Fig. 2](#) shows the normalized impact values (according to ReCiPe Midpoint (H) V1.10/Europe Recipe H method) of the bioenergy system for a selection of environmental categories, and also the contribution of each one of the four life cycle phases considered in this analysis (soil conditioning, cultivation, harvesting and soil recovery). The graph shows that the category most significantly affected by far is agricultural land occupation. The next most affected categories are the ones related to toxicity (to humans and to the marine and freshwater ecosystems) and fresh water eutrophication. Climate change and depletion of fossil resources are comparatively less significant.

3.2. Analysis of the harvesting technology: bale harvesting vs. chip harvesting

[Fig. 3](#) illustrates the environmental performance of the bioenergy system considering the two biomass collection scenarios: bale and chip harvesting. A 25 km distance between the cultivation

Table 3
Characterized impacts of the bioenergy supply system (bale harvesting and 25 km transport).

	Total		% Contribution to total impact			
	Value	Units/GJ	Soil condit.	Cultivation	Harvesting	Soil recov.
Climate change	2.33	Kg CO ₂ eq	9.7	30.9	57.3	2.1
Terrestrial acidification	23.5	g SO ₂ eq	12.5	51.1	35.0	1.4
Freshwater eutrophication	0.492	g P eq	24.5	26.1	48.8	0.6
Marine eutrophication	3.80	g N eq	14.1	73.9	11.5	0.5
Human toxicity	567	g 1,4-DB eq	12.3	26.3	60.7	0.7
Photochemical oxidant formation	17.7	g NMVOC	8.7	21.4	66.6	3.2
Particulate matter formation	7.5	g PM10 eq	10.5	32.7	54.2	2.6
Freshwater ecotoxicity	14.8	g 1,4-DB eq	11.3	24.7	63.3	0.7
Marine ecotoxicity	15.6	g 1,4-DB eq	11.4	24.6	63.1	0.8
Agricultural land occupation	29.2	m ² yr	0.03	99.8	0.004	0
Water depletion	12.5	m ³	1.7	79.2	19.0	0.1
Fossil depletion	0.601	kg oil eq	9.0	21.3	67.0	2.7

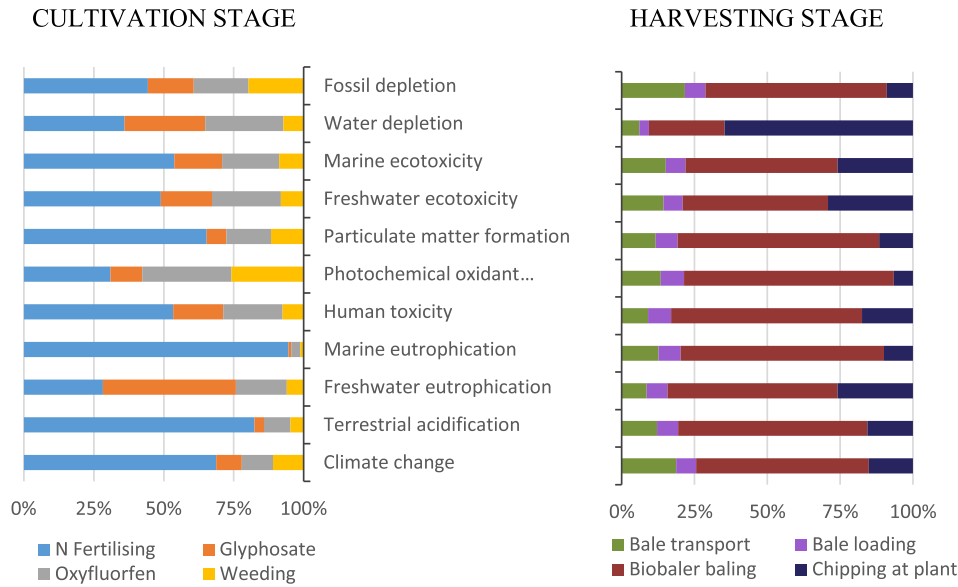


Fig. 1. Contribution of specific processes to the cultivation and harvesting phases in the bioenergy system (as % of characterized impact).

plot and the energy plant is considered in both cases. The results evidence reduced environmental impacts in most categories when using the chip harvesting technology (15% lower impact on average and up to 35% lower impact in the case of marine ecotoxicity). This is due primarily to the higher fuel requirements of the bale harvester and also to the electricity consumed for bale chipping at the energy plant. The harvesting technology has no effect on the agricultural land occupation category.

3.3. Sensitivity analysis of biomass transportation distance

Table 5 shows the effect of transportation distance between the cultivation plot and the energy plant, on the characterized impacts of the bioenergy system. The analysis is based on the bale harvesting scenario. As expected, the environmental performance of the system deteriorated with the distance in all categories. Depletion of fossil resources and climate change are the categories most significantly affected (57% and 41% increase when transportation distance changed from 10 to 100 km) due primarily to the consumption of transportation fuels.

The results show a linear relationship between distance and impact value in all midpoint indicators. Equations (1)–(4) describe

this relationship for selected categories where (x) represents distance and (y) the environmental impact value:

$$\text{Climate change: } y \text{ (kg CO}_2 \text{ eq/GJ)} = 0.0138x + 2.8718 \quad (1)$$

$$\text{Freshwater ecotoxicity: } y \text{ (g 1.4-DB eq/GJ)} = 0.0743x + 18.613 \quad (2)$$

$$\text{Marine ecotoxicity: } y \text{ (g 1.4-DB eq/GJ)} = 0.0826x + 19.531 \quad (3)$$

$$\text{Photochem. oxidant: } y \text{ (g NMVOC/GJ)} = 0.0872x + 22.3449 \quad (4)$$

The constant term in the algebraic expression represents the impact values at the cultivation plot (when transportation distance is not considered).

3.4. Energy analysis of the bioenergy supply system

Table 6 shows the energy intensity of the bioenergy system in terms of Cumulative Energy Demand (CED) and Net Energy Ratio (NER). The results obtained for the base case scenario (bale harvesting and 25 km distance to the energy plant) are characteristic of a highly efficient energy system with a CED of 29.1 MJ/GJ and a NER

Table 4
Characterized impacts related to the consumption of phytosanitary and fertilizing products.

Impact category		Total		% Contribution			
		Value	Units/GJ	Production	Transport	Application	Diffuse emissions
N fertilizing	Climate change	0.49	Kg CO ₂ eq	63	3.0	5.8	28
	Terrestrial acidification	9.90	g SO ₂ eq	17	0.70	2.1	80
	Marine eutrophication	2.65	g N eq	4	0.2	0.5	96
	Human toxicity	79.4	g 1.4-DB eq	90	3.1	7.0	0
	Marine ecotoxicity	2.07	g 1.4-DB eq	92	1.5	6.4	0
Oxyfluorfen	Climate change	0.081	kg CO ₂ eq	71	0.32	28	0
	Terrestrial acidification	1.14	g SO ₂ eq	85	0.09	15	0
	Marine eutrophication	0.09	g N eq	89	0.06	11	0
	Human toxicity	31.54	g 1.4-DB eq	87	0.28	12	0.39
	Marine ecotoxicity	0.79	g 1.4-DB eq	87	0.30	13	0.01
Glyphosate	Climate change	0.065	kg CO ₂ eq	65	0.34	35	0
	Terrestrial acidification	0.40	g SO ₂ eq	59	0.22	41	0
	Freshwater eutrophication	0.061	g P eq	96	0.07	4.2	0
	Human toxicity	26.69	g 1.4-DB eq	85	0.28	14	0.002
	Marine ecotoxicity	0.66	g 1.4-DB eq	84	0.30	15	0.008

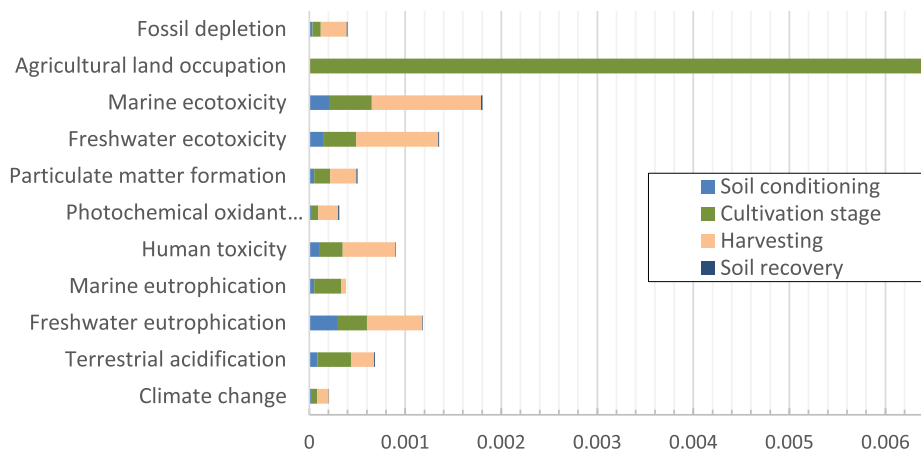


Fig. 2. Normalized analysis of the environmental performance of the bioenergy supply system (bale harvesting and 25 km).

of 34. These values are comparable to those published by other authors for similar poplar plantations (Gonzalez-Garcia et al., 2014).

Harvesting is the most energy intensive phase in the life cycle of the bioenergy system, consuming 19.8 MJ/GJ. This is followed by cultivation and soil conditioning, while the intensity of the soil recovery phase is negligible. CED values followed a polynomial function with the transportation distance (x), as shown in Equation (5).

$$\text{CED: } y \text{ (MJ/GJ)} = 0.0004x^2 + 0.0991x + 25.771 \quad (5)$$

Chip harvesting is less energy intensive than bale harvesting resulting in a comparatively lower CED (21.1 MJ/GJ) and a consequently higher NER (48). This is due primarily to the lower energy requirements of the chip harvesting process (11.7 MJ/GJ), compared to bale harvesting (19.8 MJ/GJ). The energy indicators for the other life cycle phases remain unchanged.

Fig. 4 illustrates the contribution of different operations to the CED of the harvesting phase. The results evidence that most (52%) of the energy consumed in the chip harvesting phase (11.7 MJ/GJ) correspond to construction and operation of the Claas Jaguar forager (5.90 MJ/GJ). The CED associated with the tractors and trailers represents 3.71 MJ/GJ, while transportation of the wood chips to the energy plant amounts to the remaining 2.11 MJ/GJ. Harvesting with the Biobaler™ is very energy intensive (11.5 MJ/GJ). Most of this energy (56%) is consumed directly in the

harvesting process, with a smaller contribution attributable to transportation of the biomass to the energy plant (20%) and bale loading onto the lorries (7.0%). The results in Fig. 4 also evidence that the transportation process is more energy intensive in absolute terms in the bale harvesting scenario, due to the lower bulk density of the air dried bales (compared to the humid wood chips). This means more trips from the plot to the energy plant and, therefore, more distance covered and more fuel consumed by the lorries.

3.5. Economic analysis of biomass supply chain based on poplar SRC

Table 7 illustrates the economics of the bioenergy project (including bale and chip harvesting scenarios) in terms of Net Present Value (NPV). The monetary values refer to cash inflows/outflows actualized at the end of the cultivation period (2021) when assuming an annual discount rate of 5.0% and an inflation rate of 0.0%. The results show higher operating costs in the bale harvesting scenario (−32256 €/ha, equivalent to −7.8 €/GJ) compared to chip harvesting (−29860 €/ha, equivalent to −7.3 €/GJ). Renting of the cultivation land is by far the priciest item, representing 39.2% of the actualized cash outflows in the bale harvesting scenario. Comparable rental costs have been reported by other authors for similar agricultural activities in Spain and Italy (Gasol et al., 2009a,b; Manzone et al., 2014).

Harvesting was the second most expensive item contributing to 24.5% and 20.7% of all the actualized cash outflows in the bale and

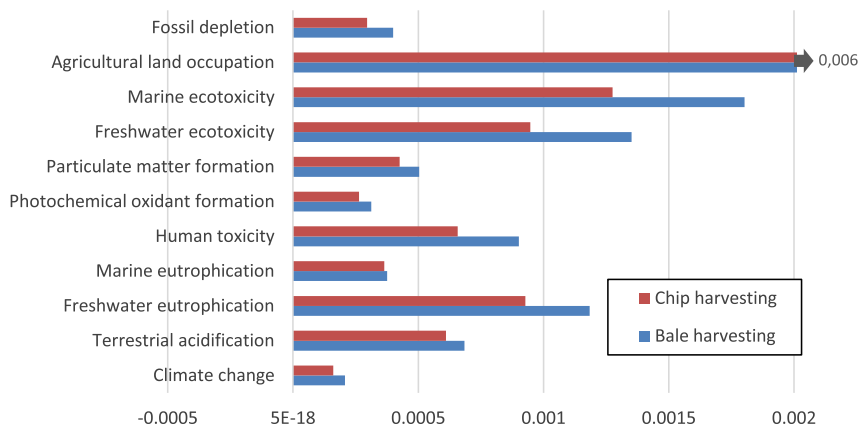


Fig. 3. Environmental performance (normalized values) of alternative harvesting technologies (bale vs chip harvesting) in the poplar SRC.

Table 5

Effect of distance on the environmental performance of the bioenergy supply system.

Impact category	Unit/FU	Transportation distance (km)						10–100 variation
		10	25	40	60	80	100	
Climate change	Kg CO ₂ eq	2.18	2.33	2.48	2.67	2.87	3.07	41%
Terrestrial acidification	g SO ₂ eq	22.9	23.5	24.1	24.9	25.7	26.5	16%
Freshwater eutrophication	g P eq	0.48	0.49	0.50	0.52	0.54	0.55	15%
Marine eutrophication	g N eq	3.77	3.80	3.84	3.88	3.93	3.97	5%
Human toxicity	g 1.4-DB eq	548	567	586	610	635	660	20%
Photochem. oxidant formation	g NMVOC	16.8	17.7	18.7	19.9	21.2	22.5	34%
Particulate matter formation	g PM10 eq	7.21	7.50	7.78	8.16	8.54	8.92	24%
Freshwater ecotoxicity	g 1.4-DB eq	14.0	14.8	15.6	16.7	17.8	18.8	35%
Marine ecotoxicity	g 1.4-DB eq	14.7	15.6	16.5	17.7	18.9	20.1	37%
Agricultural land occupation	m ² yr	29.2	29.2	29.2	29.2	29.2	29.2	0%
Water depletion	m ³	12.4	12.5	12.6	12.7	12.8	12.9	4%
Fossil depletion	kg oil eq	0.55	0.60	0.65	0.72	0.79	0.86	57%

chip harvesting scenarios, respectively. The higher disbursements observed in the bale harvesting scenario are due to the higher costs of biomass collection with the Biobaler™ and also to the higher transportation costs of the bales, which exhibit lower bulk densities than the wood chips. Irrigation also has a strong influence on the economics of the system contributing to 15.4% and 16.6% of the cash outflows (bale harvesting and chip harvesting scenarios, respectively). Biomass transportation (25 km) (8.2% and 9.7%) and other agricultural activities have a lesser influence on the economics of the system.

Regarding cash inflows, actualized revenues were slightly higher in the case of bale harvesting (5.50 €/GJ) compared to chip harvesting (5.16 €/GJ) due to the lower moisture content and higher market value of the wood chips produced in the former scenario (60 €/t vs 45 €/t). Under the assumptions described, the NPV of the bioenergy supply chain is unfavorable in both scenarios (−9613 €/ha or −2.33 €/GJ in the case of bale harvesting and −8631 €/ha or −2.10 €/GJ with chip harvesting). The higher revenues achieved with bale harvesting do not compensate the higher cost incurred. Hence, chip harvesting appears as a better option from an economic stand point, although the difference is not significant.

Fig. 5 illustrates the economic viability of the system (in terms of NPV and IRR) as a function of wood chip prices. The results show that NPV equals zero when wood chips sell for a current market price of 5.88 €/GJ in the bale harvesting scenario and 5.79 €/GJ in the chip harvesting scenario. This is equivalent to 63.3 €/t of chip harvested wood chips (40 wt% moisture) or 84.4 €/t of bale harvested wood chips (25 wt% moisture).

Fig. 6 describes the influence of transportation distance on the economics of the bioenergy system. In the bale harvesting scenario, transportation costs for a distance of 10 km from the cultivation plot to the energy plant amount to 0.29 €/MJ, representing 4.8% of the cash outflows of the system. This value rises to 1.33 €/MJ when the transportation distance is 100 km, representing 19.9% of all the cash outflows. As shown in Fig. 6, transportation costs are

comparatively lower in the case of chip harvesting due to the higher bulk density of this product.

Under the conditions described, the biomass supply chain based on poplar SRC is not economically viable. In order to improve the economics of the system, the following elements may be considered: reduce irrigation costs by implementing this type of crops in locations with higher rainfall rates; increase biomass productivity by optimizing cultivation strategies and technologies; minimize transportation costs; and selection of cultivation location to reduce land rental costs. The application of public subsidies for the production of biomass fuels or indirect subsidies to activities consuming biomass fuels (like power generation, domestic and industrial heat applications), as it happens in other European countries like Germany, Sweden or UK, would also improve the economics of the system (Faasch and Patenaude, 2012).

4. Conclusions

- The carbon footprint of a biomass supply chain based on a poplar plantation cultivated as a SRC in Southern Spain has been calculated to be 2.33 kg CO₂eq/GJ (assuming bale harvesting and 25 km distance to the energy plant). Impacts on other key categories were as follows: freshwater eutrophication (0.492 g P eq/GJ), human toxicity (567 g 1.4-DB eq/GJ), terrestrial

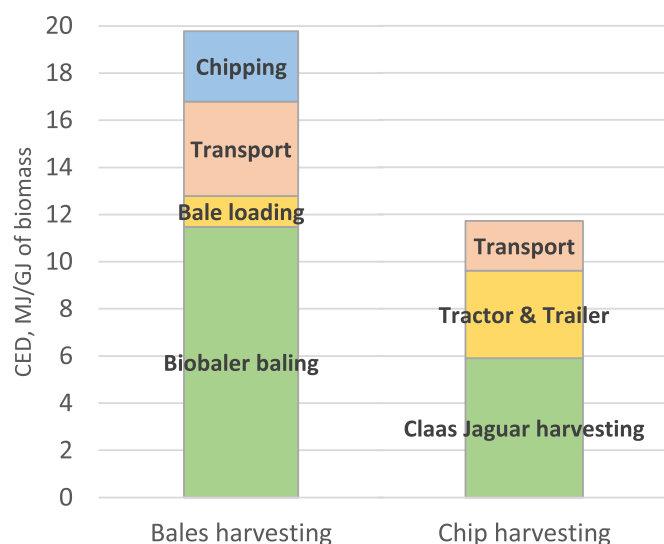


Fig. 4. Contribution of different processes to the CED of alternative biomass collection phases using bale and chip harvesting technologies.

Table 6

Energy efficiency (CED and NER) of the bioenergy supply system for two biomass collection scenarios (bale and chip harvesting).

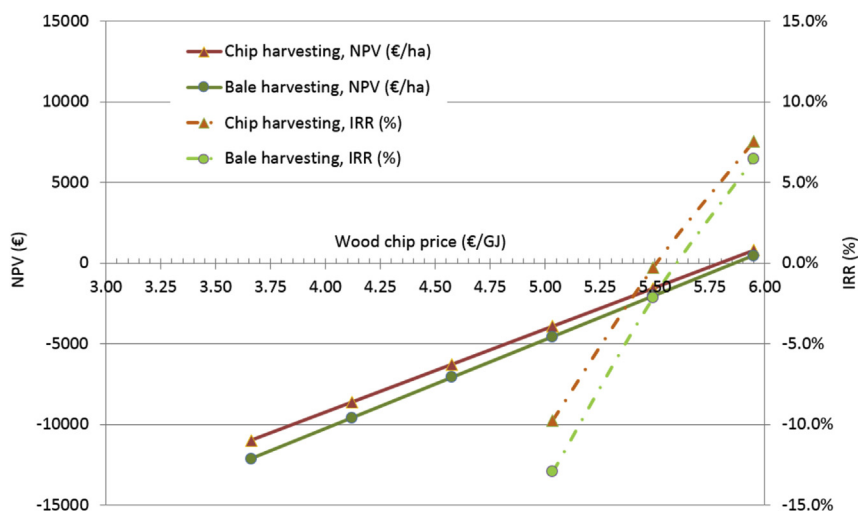
	Total	Soil conditioning	Cultivation	Harvesting	Soil recovery	NER
Bale harvesting	29.1	2.5	6.1	19.8	0.7	34
Chip harvesting	21.1	2.5	6.1	11.7	0.7	48

Table 7

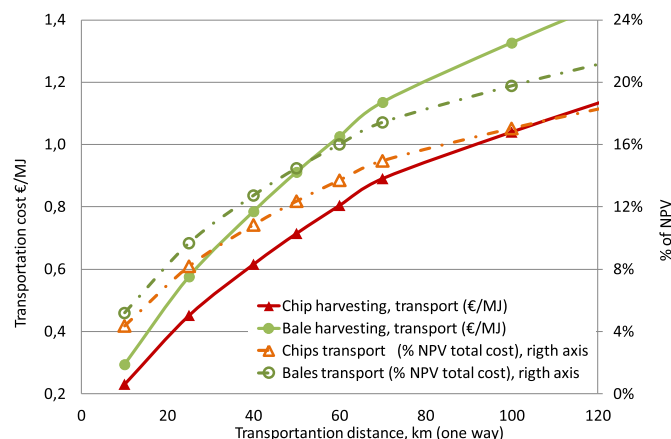
Economic analysis of the bioenergy supply chain including two biomass collection scenarios: bale and chip harvesting.

	Bale harvesting scenario			Chip harvesting scenario		
	€/ha	€/GJ	Share of cash outflows (%)	€/ha	€/GJ	Share of cash outflows (%)
Land rental	–12651	–3.07	39.2	–12651	–3.07	42.4
Soil conditioning (subsoiling, NPK fert & surface plowing)	–659	–0.16	2.0	–659	–0.16	2.2
Mechanical weed control	–294	–0.07	0.9	–294	–0.07	1.0
Pre-emergence herbicide (oxyfluorfen)	–298	–0.07	0.9	–298	–0.07	1.0
Irrigation rights (watering cost)	–4966	–1.21	15.4	–4966	–1.21	16.6
Planting (planter for 13,333 cuttings/ha)	–1167	–0.28	3.6	–1167	–0.28	3.9
Post-emergence herbicide (glyphosate)	–137	–0.03	0.4	–137	–0.03	0.5
Harvesting (including chipping of round bales at plant)	–7914	–1.92	24.5	–6192	–1.50	20.7
Transportation (25 km distance, one way)	–3122	–0.76	9.7	–2448	–0.59	8.2
Fertilization (N fertilizer)	–575	–0.14	1.8	–575	–0.14	1.9
Land restoration (rising of stumps)	–473	–0.11	1.5	–473	–0.11	1.6
OVERALL OPERATING COSTS	–32256	–7.80	100	–29860	–7.30	100
OVERALL INCOMES: selling of wood chips	22,644	5.50	–	21,229	5.16	–
BENEFITS (NPV)	–9613	–2.33		–8631	–2.10	

Cash flows and NPV values actualized at the end of the 12 year cultivation period (2021) assuming a 5.0% discount rate and 0.0% inflation.

**Fig. 5.** Economic analysis of the bioenergy supply system (NPV and IRR) for different wood chip market prices (25 km distance).

acidification (23.5 g SO₂ eq/GJ) and agricultural land occupation (29.2 m²/yr). The Cumulative Energy Demand (CED) and Net Energy Ratio (NER) of the system were 29.1 MJ/GJ and 34, respectively.

**Fig. 6.** Transportation costs calculated for different distances between the cultivation plot and the energy plant.

- Harvesting was the most environmentally harmful phase of the system due primarily to the high energy intensity of the Bio-balerTM forager. The cultivation phase had a significant effect primarily on water depletion due to direct water use for irrigation.
- Environmental damage associated with the consumption of synthetic N-fertilizers and herbicides (glyphosate and oxyfluorfen) was associated primarily with the fabrication of these chemicals and to a lesser extent to their application and diffusion into the environment.
- Poplar collection by chip harvesting was significantly more energy efficient (CED = 21.1 MJ/GJ and NER = 48), more benign to the environment (15% lower average impact) and more economically attractive than bale harvesting.
- Biomass production costs (actualized at the end of the cultivation project) were calculated to be 7.8 €/GJ and 7.3 €/GJ in the bale and chip harvesting scenarios, respectively. Actualized cash inflows from selling the wood chips amount to 5.50 €/GJ and 5.16 €/GJ, respectively, which are insufficient to make the project economically viable.
- Biomass production costs are primarily associated with land rental (39%), harvesting (21%) and transportation (8% considering 25 km). Despite the detrimental environmental consequences associated with the use of fertilizers and herbicides,

these chemicals had a very limited contribution to the economics of the system.

- The following actions are recommended in order to improve the economic viability: reduce irrigation costs by implementing these crops in locations with higher rainfall rates; increase biomass productivity by optimizing cultivation strategies and technologies; minimize harvesting and transportation costs using chip harvesting; and select locations with reduced land rental costs. The application of public subsidies for the production of biomass fuels is necessary to achieve the economic viability.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jclepro.2015.01.070>.

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